Day 1

1. What is the OSI Model?

The **OSI (Open Systems Interconnection) model** is a conceptual framework that explains how data travels between devices over a network.

- Developed by the **ISO (International Organization for Standardization)**.
- Contains **7 layers**.
- Each layer performs a specific networking function.
- Helps understand, design, and troubleshoot networks.

2. Seven Layers of the OSI Model

Layer	Name	Main Function	Examples
7	Application	Provides network services to applications	HTTP, HTTPS, DNS, SMTP
6	Presentation	Data formatting, encryption, compression	TLS/SSL, JPEG, ASCII
5	Session	Establishes, manages, and terminates sessions	Session management
4	Transport	End-to-end delivery, reliability, and port numbers	TCP, UDP
3	Network	Logical addressing and routing	IP, routers
2	Data Link	Frames, MAC addresses, switching	Ethernet, switches
1	Physical	Transmits raw bits through physical media	Cables, hubs, radio signals

Mnemonic (Layer 7 → Layer 1):

All People Seem To Need Data Processing.

- A — Application
- P — Presentation
- S — Session
- T — Transport
- N — Network
- D — Data Link
- P — Physical

3. Important Concepts

Layer 1: Physical

- Transmits data as electrical, optical, or radio signals.
- Deals with cables, connectors, and signal transmission.
- PDU: Bits.

Layer 2: Data Link

- Uses MAC addresses for communication within a local network.
- Detects transmission errors at the frame level.
- Switches primarily operate at this layer.
- PDU: Frames.

Layer 3: Network

- Uses IP addresses to identify networks and hosts.
- Determines the path for packets between networks.
- Routers primarily operate at this layer.
- PDU: Packets.

Layer 4: Transport

- Provides end-to-end communication.
- TCP provides reliable, ordered delivery.
- UDP provides connectionless delivery with lower overhead.
- Uses port numbers to identify applications.

- PDU: Segments (TCP) or datagrams (UDP).

Layer 5: Session

- Establishes, maintains, and terminates communication sessions.

Layer 6: Presentation

- Handles data translation, encryption/decryption, and compression.

Layer 7: Application

- Provides network services to applications.
- Examples: HTTP for web browsing and SMTP for email transmission.

4. PDU – Protocol Data Units

A **PDU** is the name given to data at different OSI layers.

Layer	**PDU**
Application (7)	Data
Presentation (6)	Data
Session (5)	Data
Transport (4)	Segment / Datagram
Network (3)	Packet
Data Link (2)	Frame
Physical (1)	Bits

Remember: Data → Segment → Packet → Frame → Bits.

5. Encapsulation and De-encapsulation

Encapsulation: When *sending data*, each layer adds its own information, such as headers, as the data moves from Layer 7 down to Layer 1.

Example:

`Data → Segment → Packet → Frame → Bits`

De-encapsulation: When *receiving data*, the receiving device processes the information from Layer 1 up to Layer 7, removing the relevant headers.

`Bits → Frame → Packet → Segment → Data`

6. OSI Model vs. TCP/IP Model

OSI Model	**TCP/IP Model**
7 layers	Usually 4 layers
Conceptual reference model	Practical networking model
Separates Session and Presentation layers	Includes their functions in the Application layer
Separates Data Link and Physical layers	Combines them into the Network Access layer

7. Important Exam Points for Network+

- **Router:** Primarily Layer 3.
- **Switch:** Primarily Layer 2.
- **Hub:** Layer 1.
- **MAC address:** Layer 2.
- **IP address:** Layer 3.
- **Port numbers:** Layer 4.

- **TCP and UDP:** Layer 4.
- **HTTP, HTTPS, DNS, SMTP:** Layer 7 protocols.
- **Encryption and compression:** Commonly associated with Layer 6.
- **Frames:** Layer 2.
- **Packets:** Layer 3.
- **Segments:** Layer 4 for TCP.

Exam tip: Devices can operate across multiple layers, so remember the *primary* layer associated with each device.

Quick Revision

Question	Answer
How many OSI layers are there?	7
Which layer handles routing?	Layer 3
Which layer uses MAC addresses?	Layer 2
Which layer uses port numbers?	Layer 4
Which layer transmits bits?	Layer 1
Which layer handles encryption in the OSI model?	Layer 6
Which layer provides services to applications?	Layer 7
Which layer is associated with TCP and UDP?	Layer 4